5.9 Derivatives or Integrals of a Chebyshev-Approximated Function

If you have obtained the Chebyshev coefficients that approximate a function in a certain range (e.g., from `chebft` in §5.8), then it is a simple matter to transform them to Chebyshev coefficients corresponding to the derivative or integral of the function. Having done this, you can evaluate the derivative or integral just as if it were a function that you had Chebyshev-fitted *ab initio*.

The relevant formulas are these: If c_i, $i = 0, \ldots, m-1$ are the coefficients that approximate a function f in equation (5.8.9), C_i are the coefficients that approximate the indefinite integral of f, and c'_i are the coefficients that approximate the derivative of f, then

$$C_i = \frac{c_{i-1} - c_{i+1}}{2i} \qquad (i > 0) \tag{5.9.1}$$

$$c'_{i-1} = c'_{i+1} + 2i\,c_i \qquad (i = m-1, m-2, \ldots, 1) \tag{5.9.2}$$

Equation (5.9.1) is augmented by an arbitrary choice of C_0, corresponding to an arbitrary constant of integration. Equation (5.9.2), which is a recurrence, is started with the values $c'_m = c'_{m-1} = 0$, corresponding to no information about the $m+1$st Chebyshev coefficient of the original function f.

Here are routines for implementing equations (5.9.1) and (5.9.2). Each returns a new `Chebyshev` object on which you can `setm`, call `eval`, or use directly as a functor.

chebyshev.h

`Chebyshev Chebyshev::derivative()`

Return a new `Chebyshev` object that approximates the derivative of the existing function over the same range [a,b].

```
{
    Int j;
    Doub con;
    VecDoub cder(n);
    cder[n-1]=0.0;                          n-1 and n-2 are special cases.
    cder[n-2]=2*(n-1)*c[n-1];
    for (j=n-2;j>0;j--)                     Equation (5.9.2).
        cder[j-1]=cder[j+1]+2*j*c[j];
    con=2.0/(b-a);
    for (j=0;j<n;j++) cder[j] *= con;       Normalize to the interval b-a.
    return Chebyshev(cder,a,b);
}
```

chebyshev.h

`Chebyshev Chebyshev::integral()`

Return a new `Chebyshev` object that approximates the indefinite integral of the existing function over the same range [a,b]. The constant of integration is set so that the integral vanishes at a.

```
{
    Int j;
    Doub sum=0.0,fac=1.0,con;
    VecDoub cint(n);
    con=0.25*(b-a);                         Factor that normalizes to the interval b-a.
    for (j=1;j<n-1;j++) {
        cint[j]=con*(c[j-1]-c[j+1])/j;      Equation (5.9.1).
        sum += fac*cint[j];                 Accumulates the constant of integration.
        fac = -fac;                         Will equal ±1.
    }
    cint[n-1]=con*c[n-2]/(n-1);             Special case of (5.9.1) for n-1.
```